9C Lesson 7

In and out: surface area, diffusion and osmosis

Year 9 Science · 9C Threat from disease · Student Book pages 50 to 52

Name _____ Class _____ Date _____

Today I can:

☐ I can calculate a surface area : volume ratio and explain why it matters for cells and organisms.

☐ I can describe how osmosis happens through a partially permeable membrane and use it to explain what I see.

Key words

surface area the total area of the outside of an object

volume the amount of space an object takes up

ratio one quantity compared with another, e.g. 6 : 1

diffusion the overall movement of particles from where they are more concentrated to where they are less concentrated

concentration gradient a difference in concentration between two places

osmosis the diffusion of water molecules through a partially permeable membrane

partially permeable membrane a membrane with tiny holes that let small molecules through but not large ones

cell membrane the partially permeable layer around a cell

unicellular made of one cell

villi finger-like folds that give the small intestine a large surface area

pandemic an outbreak of a disease across the whole world

● Core Everyone does this section.

1. Complete the text using words from the word bank.

The movement of all particles is _____. Where there is a concentration _____, more particles move out from a region of _____ concentration than from a region of _____ concentration. This spreading out is known as _____.

2. Complete the table for three cubes.

Cube	Surface area (cm2)	Volume (cm3)	Surface area : volume ratio
sides 1 cm	6	1	6 : 1
sides 2 cm		8	
sides 3 cm	54		

3. Potato strips in water and salt solution.

a) Record your results.

Strip	Liquid	Length at start (mm)	Length at end (mm)	Change (mm)	Firm or floppy?
A	pure water				
B	salt solution				

b) Tick the strip that gained water.

☐ A strip A (water)

☐ B strip B (salt solution)

c) Explain your results using the word osmosis.

●● Stretch When Core is finished.

S1. A cuboid has length 3 cm, width 2 cm and height 1 cm. Surface area = 2(l x h) + 2(l x w) + 2(w x h). Volume = l x w x h.

a) Calculate the surface area : volume ratio of this cuboid. Show all your working.

b) Compare it with the surface area : volume ratio of a cuboid with l = 6 cm, w = 4 cm and h = 2 cm.

S2. Oxygen diffuses out of the stomata of a leaf during the day.

a) Explain why oxygen diffuses out of a leaf more rapidly on a hot day than on a cold day.

b) Explain why oxygen diffuses out of a leaf more rapidly on a windy day than on a still day. (Hint: think about the concentration gradient.)

●●● Challenge Explain in full sentences.

C1. A silky anteater has a mass of about 400 g and eats up to 5000 ants a day. A giant anteater has a mass of about 40 kg and eats up to 30 000 ants a day.

a) Calculate the number of ants eaten per gram of body mass for each anteater.

b) Suggest a reason for the difference, using ideas about surface area : volume ratio.

C2. Osmosis in living things.

a) Paramecium is a unicellular organism that lives in fresh water. It has a contractile vacuole that collects water from the cytoplasm and pumps it out. Explain why Paramecium needs to remove water from its cytoplasm.

b) Tropical sprue stops food being digested properly, so undigested food stays dissolved inside the small intestine. Explain how this can cause diarrhoea (watery faeces) by osmosis.

My thinking (any language)

An elephant and a mouse: which one loses heat faster for its size, and how does this affect how much each must eat? Use the words surface area and volume.

Word bank random · gradient · higher · lower · diffusion · osmosis · partially permeable · surface area · volume · ratio

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Exit ticket 9C Lesson 7: In and out: surface area, diffusion and osmosis Name _____

1. What is osmosis?

2. A 2 cm cube has a surface area of 24 cm2 and a volume of 8 cm3. State its surface area : volume ratio.

3. Give one reason why rapid international travel makes a pandemic more likely.

Today I can: Goal 1 ☐ Goal 2 ☐